

What is this "Jamming" I hear about?

Jamming is sort of a Denial of Service (DoS) attack against your wireless network. It can be achieved in many ways by an attacker. The easiest method is to send out a much stronger signal than your access point, using the same SSID and channel of the 2.4 GHz frequency that your network uses. This will confuse all your wireless clients, and end up reducing your network bandwidth, or even killing your network. Use one of the tools mentioned in Chapter 5 to survey your premises and find out whether there are any rogue devices in the vicinity. If there are, change your SSID, channel settings and WEP key immediately.

Why do I get such a low network bandwidth?

Before setting up your network you should run an amateur site survey using the tools mentioned in the previous chapter, Security, of this book. Not only do they help improve your access point placement, they also inform you of any rogue devices or interfering devices in the vicinity. If you place your access point or wireless device too close to a microwave or cordless phone that runs on the 2.4 GHz frequency, be prepared to have a lot of interference, and perhaps even network data loss. Make sure to keep other devices that run at frequencies greater than 2 GHz as far away as possible from your wireless access points and clients.

What are the real-life data transfer speeds I can expect from a WLAN?

The 802.11b standard offers a theoretical data transfer speed of 11 Mbps (that's in bits, not bytes), and the 802.11g standard boasts of theoretical speeds of up to 54 Mbps. Though this is still very slow when compared to the 100 Mbps bandwidth offered by standard Ethernet, it would still be sufficient for most users. Of course, just as we never really get 100 Mbps on a wired LAN, you will never get the rated theoretical speeds on a WLAN either. Expect to see speeds of a maximum of 5 Mbps for 802.11b and 20 Mbps for a 802.11g network.

I have 802.11g compatible equipment, but I still don't get more than 3 or 4 Mbps transfer speeds. Why is that?